



Incorporating AI into Teaching

Kerry Back
Rice University
FMA 2025



JPMorgan: Chatbots to Agents

CNBC, September 30, 2025: In 2023, JPMorgan gave employees access to OpenAI's models through LLM Suite.

It was essentially a corporate ChatGPT tool used to draft emails and summarize documents.

About 250,000 JPMorgan employees have access to the platform today ... Half of them use it roughly every day.

JPMorgan is now early in the next phase of its AI blueprint: It has begun deploying agentic AI to handle complex multistep tasks for employees, according to an internal road map provided by the bank.

From Makers to Checkers

Derek Waldron, JP Morgan Chief Analytics Officer:

What we're working towards is that every employee will have their own personalized AI assistant; every process is powered by AI agents, and every client experience has an AI concierge.

You'll still have people at the top who are managing and have relationships with clients, but many, many of the processes underneath are now being done by AI systems.

Workers would shift from being creators of reports or software updates, or 'makers' ... to 'checkers' or managers of AI agents doing that work.

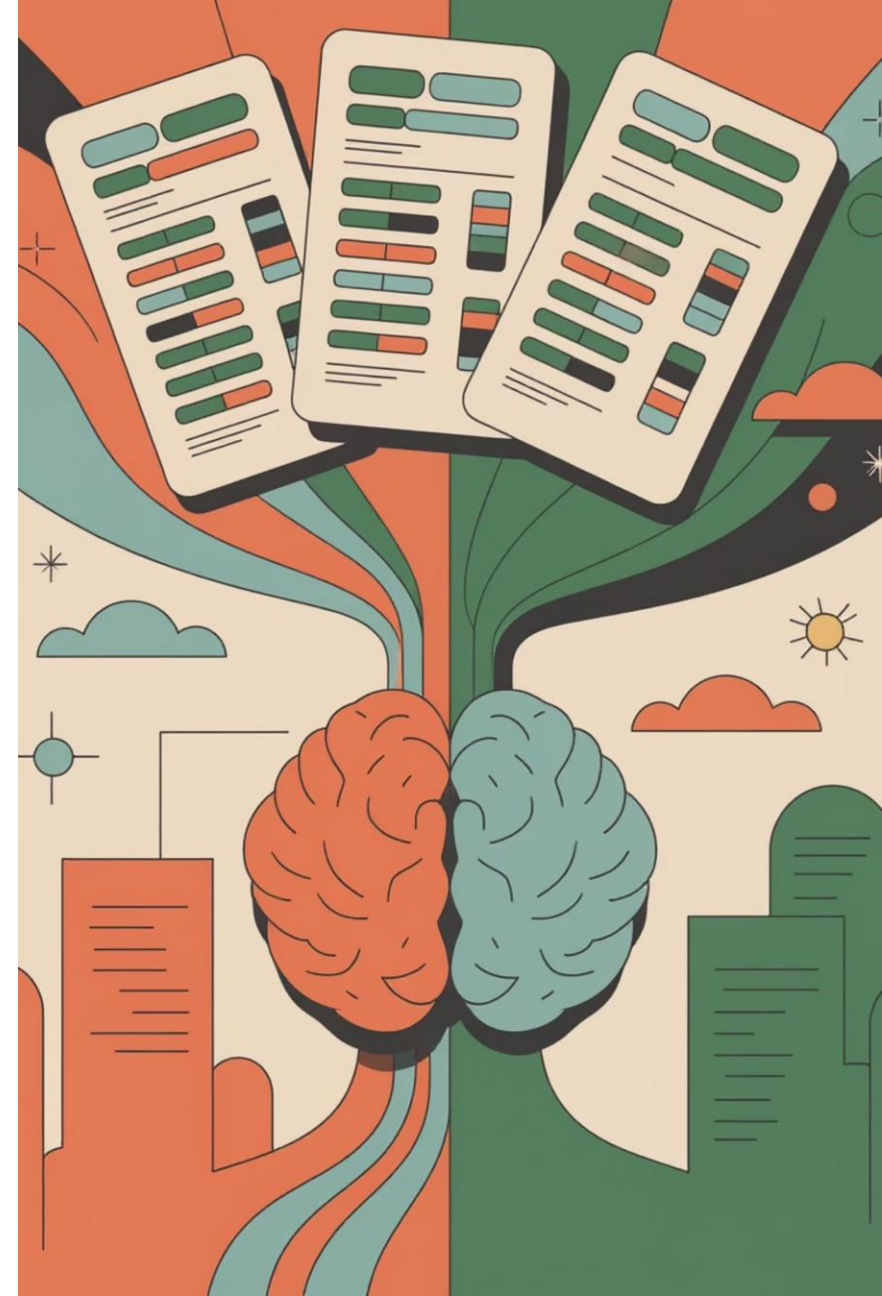
OpenAI Hires Ex Investment Bankers

Bloomberg 10-21-2025:

OpenAI has initiated a new program contracting approximately 100 former junior investment bankers.

They are tasked with creating detailed prompts and spreadsheet model responses.

OpenAI wants to get calculations, layouts, and formatting exactly right.



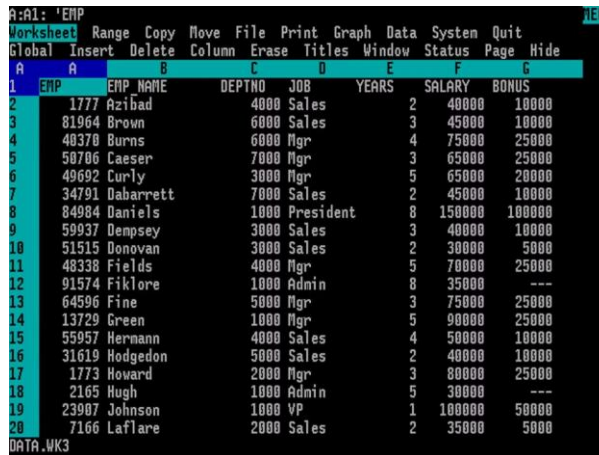
Change was only at the margins for the past 40 years



VisiCalc (1979) screenshot showing a spreadsheet with columns: ITEM, NO., UNIT, COST. The data includes items like MUCK RAKE, BUZZ CUT, TONER, and EYE SNUFF. A subtotal and tax calculation are shown at the bottom.

ITEM	NO.	UNIT	COST
MUCK RAKE	43	12.95	556.85
BUZZ CUT	15	6.75	101.25
TONER	250	49.95	12487.50
EYE SNUFF	2	4.95	9.90
SUBTOTAL			13155.50
9.75% TAX			1282.66
TOTAL			14438.16

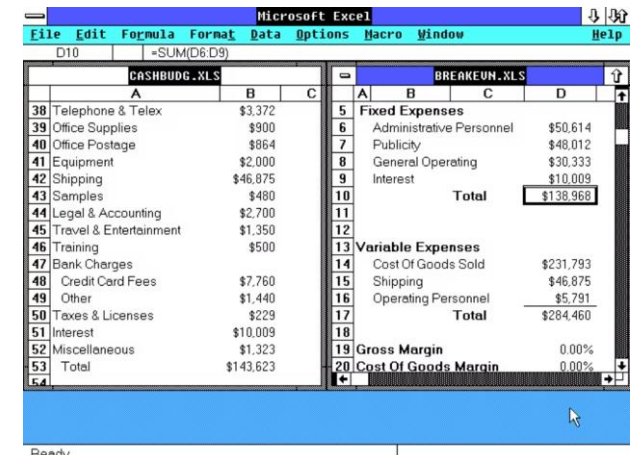
VisiCalc (1979)



Lotus 1-2-3 (1983) screenshot showing a spreadsheet with columns: EMP, EMP NAME, DEPTNO, JOB, YEARS, SALARY, BONUS. The data lists employees like Azibad, Brown, Burns, Caesar, etc., with their respective salaries and bonuses.

EMP	EMP NAME	DEPTNO	JOB	YEARS	SALARY	BONUS
1777	Azibad	4000	Sales	2	40000	10000
81964	Brown	6000	Sales	3	45000	10000
40370	Burns	6000	Mgr	4	75000	25000
50706	Caesar	7000	Mgr	3	65000	25000
49692	Curly	3000	Mgr	5	65000	20000
34791	Dabarrett	7000	Sales	2	45000	10000
84984	Daniels	1000	President	8	150000	100000
59937	Dempsey	3000	Sales	3	40000	10000
51515	Donovan	3000	Sales	2	30000	5000
48338	Fields	4000	Mgr	5	70000	25000
91574	Fiklore	1000	Admin	8	35000	---
64596	Fine	5000	Mgr	3	75000	25000
13729	Green	1000	Mgr	5	90000	25000
55957	Hermann	4000	Sales	4	50000	10000
31619	Hodgedon	5000	Sales	2	40000	10000
1773	Howard	2000	Mgr	3	80000	25000
2165	Hugh	1000	Admin	5	30000	---
23907	Johnson	1000	VP	1	100000	50000
7166	Laflare	2000	Sales	2	35000	5000

Lotus 1-2-3 (1983)



Microsoft Excel (1987) screenshot showing two spreadsheets: CASHBUDG.XLS and BREAKUN.XLS. CASHBUDG.XLS lists expenses like Telephone & Telex, Office Supplies, etc. BREAKUN.XLS shows Fixed Expenses, Variable Expenses, and Gross Margin.

CASHBUDG.XLS		
A	B	C
38	Telephone & Telex	\$3,372
39	Office Supplies	\$900
40	Office Postage	\$864
41	Equipment	\$2,000
42	Shipping	\$46,875
43	Samples	\$480
44	Legal & Accounting	\$2,700
45	Travel & Entertainment	\$1,350
46	Training	\$500
47	Bank Charges	
48	Credit Card Fees	\$7,760
49	Other	\$1,440
50	Taxes & Licenses	\$229
51	Interest	\$10,009
52	Miscellaneous	\$1,323
53	Total	\$143,623

BREAKUN.XLS		
A	B	C
5	Fixed Expenses	
6	Administrative Personnel	\$50,614
7	Publicity	\$48,012
8	General Operating	\$30,333
9	Interest	\$10,009
10	Total	\$138,968
13	Variable Expenses	
14	Cost Of Goods Sold	\$231,793
15	Shipping	\$46,875
16	Operating Personnel	\$5,791
17	Total	\$284,460
19	Gross Margin	0.00%
20	Cost Of Goods Margin	0.00%

Excel (1987)

For 40 years, the fundamental tool of business analysis remained essentially the same.



Now We're All Learning Together

AI has created an unprecedented situation:

Faculty, students, and industry professionals are all navigating the technological shift simultaneously.

Roadmap for Today's Discussion

A Course on Gen AI for Finance

Chatbots and Agents

Previous Disruptive Technologies

AI in Finance Courses

AI as a Teaching Partner

Deep Dive into Claude Code





A Course on Gen AI for Finance

Core Themes

Chatting:

- Treat AI as a colleague: collaborate, evaluate, iterate
- Coding: LLMs are unreliable for arithmetic; coding is essential

Building:

- Apps: to reuse tested code
- Custom chatbots: system prompts, RAG, or fine tuning
- Agents: chatbots with tools

Why Teach How to Build Things?



Students gain perspective on how things fit together in the AI space



Students may end up at small or medium size firms and be builders



Building apps/chatbots/agents to do finance can deepen understanding of finance



May check a box for recruiters

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May check a box for recruiters



It's fun!



Prompt Engineering

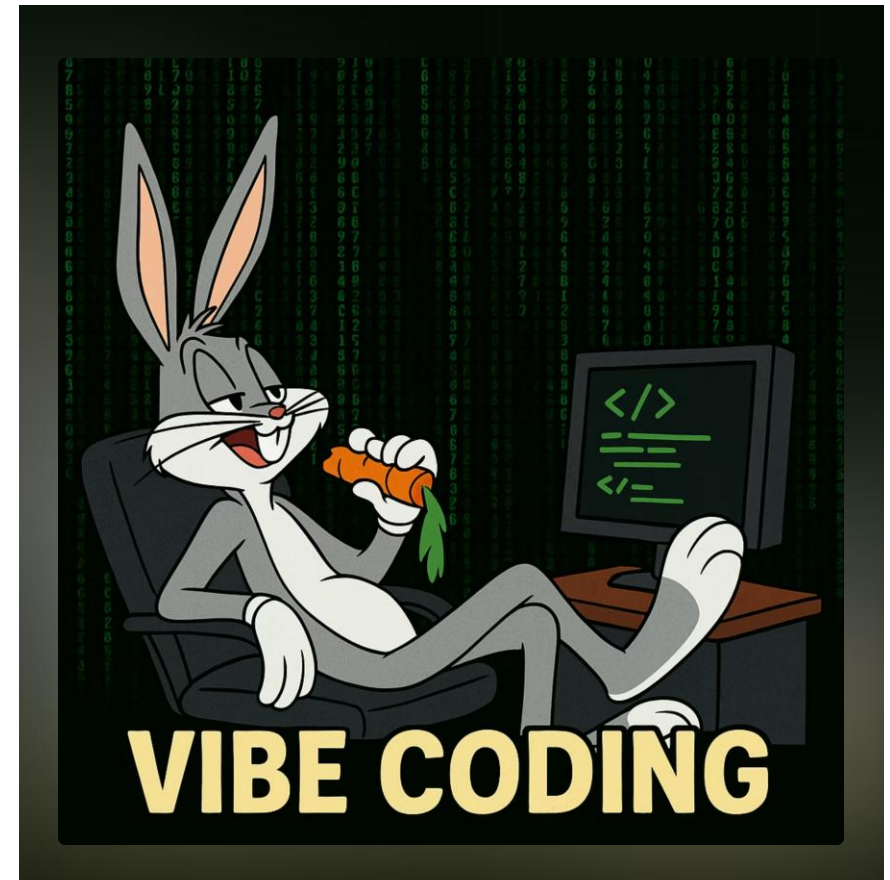
Ethan Mollick, Wharton:

Treat AI like an **infinitely patient new coworker** ... As it is a coworker, you want to work with it, not just give it orders, and you also want to learn what it is good or bad at ...
Working with AI is a dialogue, not an order.

Using AI to Write Code

Y Combinator CEO Garry Tan and Managing Partner Jared Friedman, March 2025:

For roughly a quarter of the startups in our Winter 2025 cohort, **95% of the codebase was written by AI.**





Claude + Python at Bridgewater

Aaron Linsky, CTO, AIA Labs at Bridgewater:

We've been developing capabilities powered by Claude since 2023 within AIA Labs.

Claude powered the first versions of our Investment Analyst Assistant, which streamlined our analysts' workflow by generating **Python code**, creating **data visualizations**, and iterating through **complex financial analysis tasks** with the precision of a junior analyst.

Coding Tools

Chatbots with Code Execution

ChatGPT, Gemini, Claude, **Julius.ai** - integrated environments for coding and testing

Python Environments with AI

Google Colab, VS Code, Cursor, Windsurf - traditional development enhanced by AI assistance

All-in-One Solutions

Replit - complete development and hosting platform with AI integration

My Current Preference

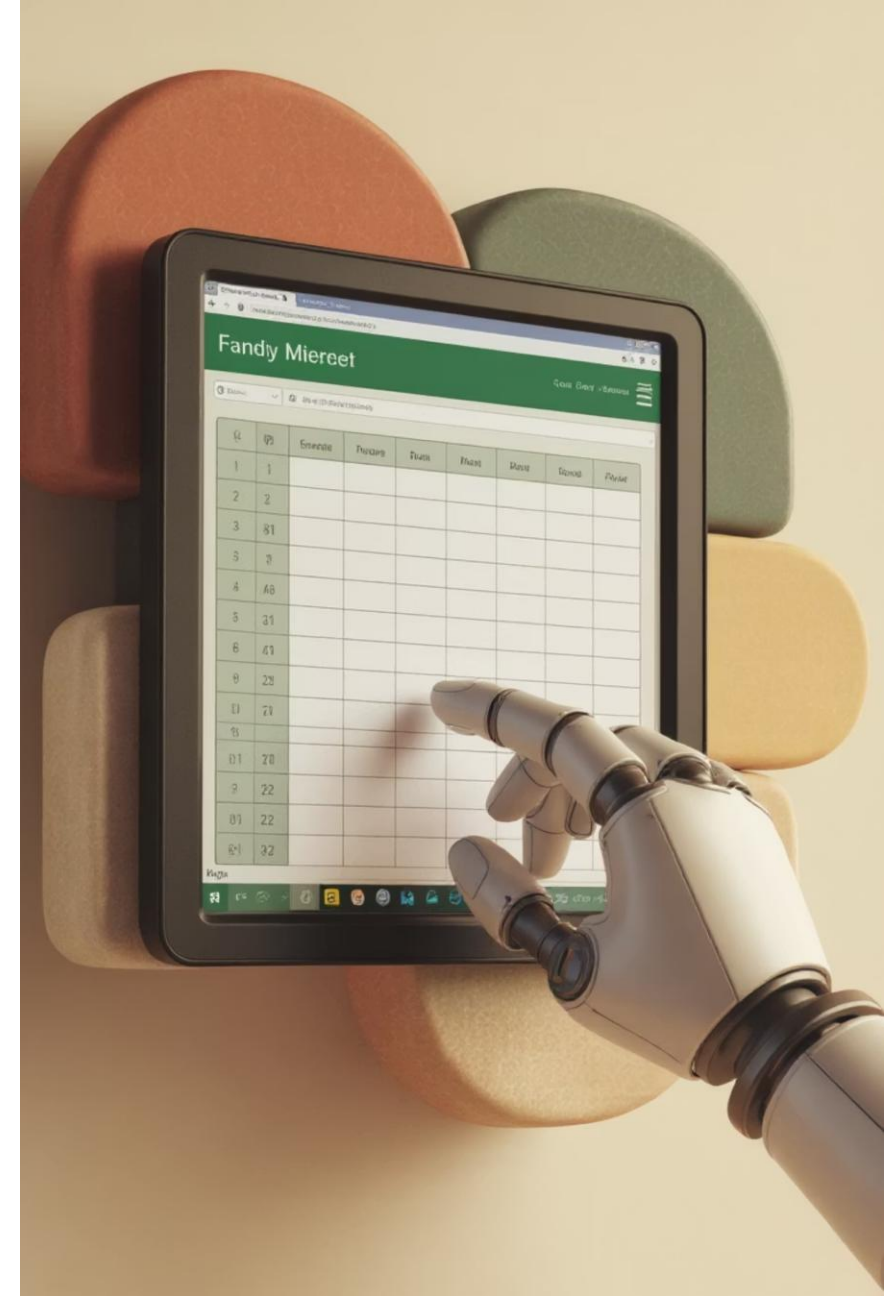
Claude Desktop (cloud Python) and **Claude Code** (local Python)

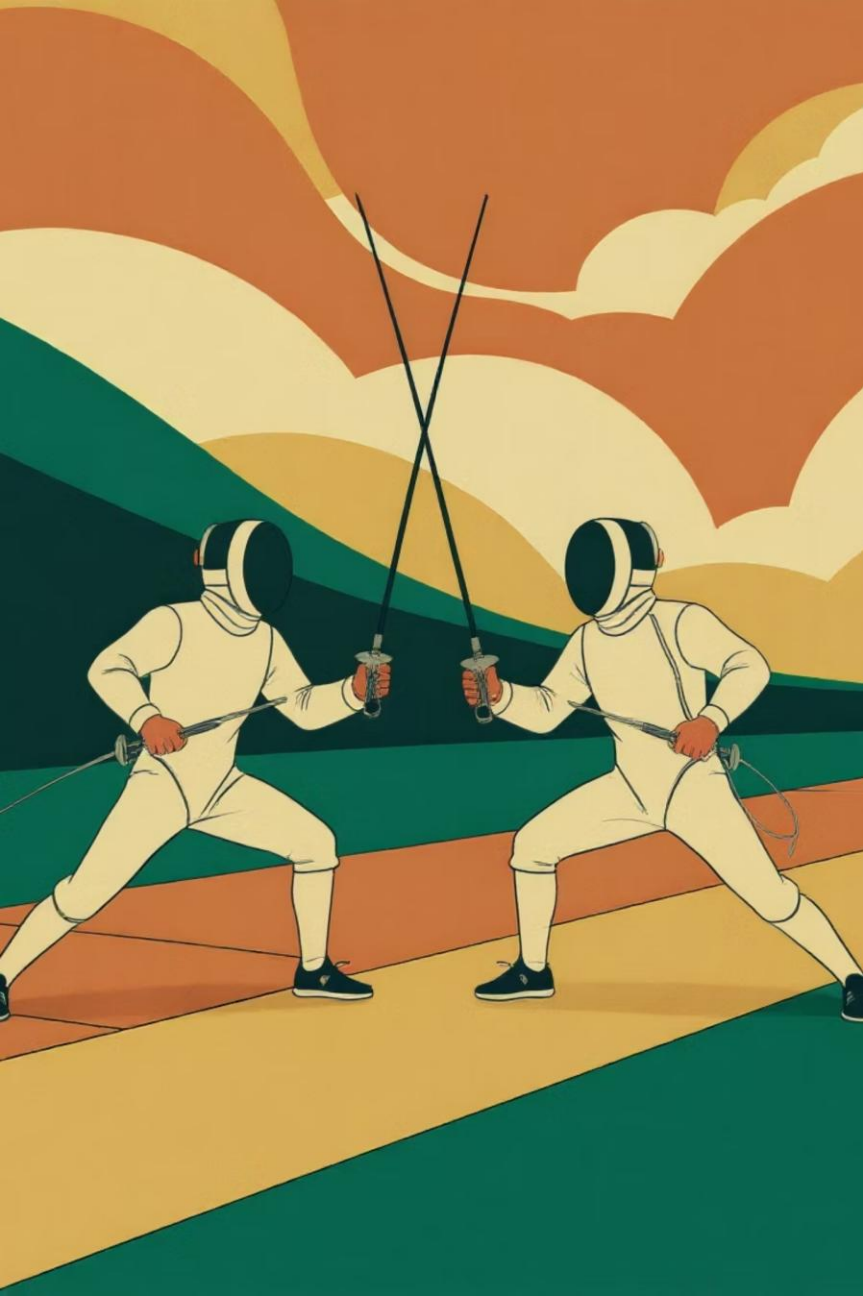
AI Can Create Excel

Claude became able to generate **fully functioning, nicely formatted Excel workbooks** in September 2025.

Microsoft has introduced the same Anthropic technology into Excel itself.

ChatGPT can also generate Excel files, but Claude seems to do a superior job at this time.





Claude Skills

- Claude can generate Excel files because of **Anthropic's new Excel Skill**.
- Recently introduced (10-16-2025), Claude Skills are customized prompts plus auxiliary materials such as code scripts.
- Anthropic has created a number of Skills, **even a Skill for creating Skills**. Users can easily create their own (ask Claude to do it).
- Claude consults Skills as needed.

Claude Excel Example

HBS Case *Valuing Wal-Mart 2010*:



Read the uploaded case. Ignore the valuation method described in the case. Do a two-stage DCF valuation. Create a PowerPoint deck and fully functioning and professionally formatted Excel workbook.

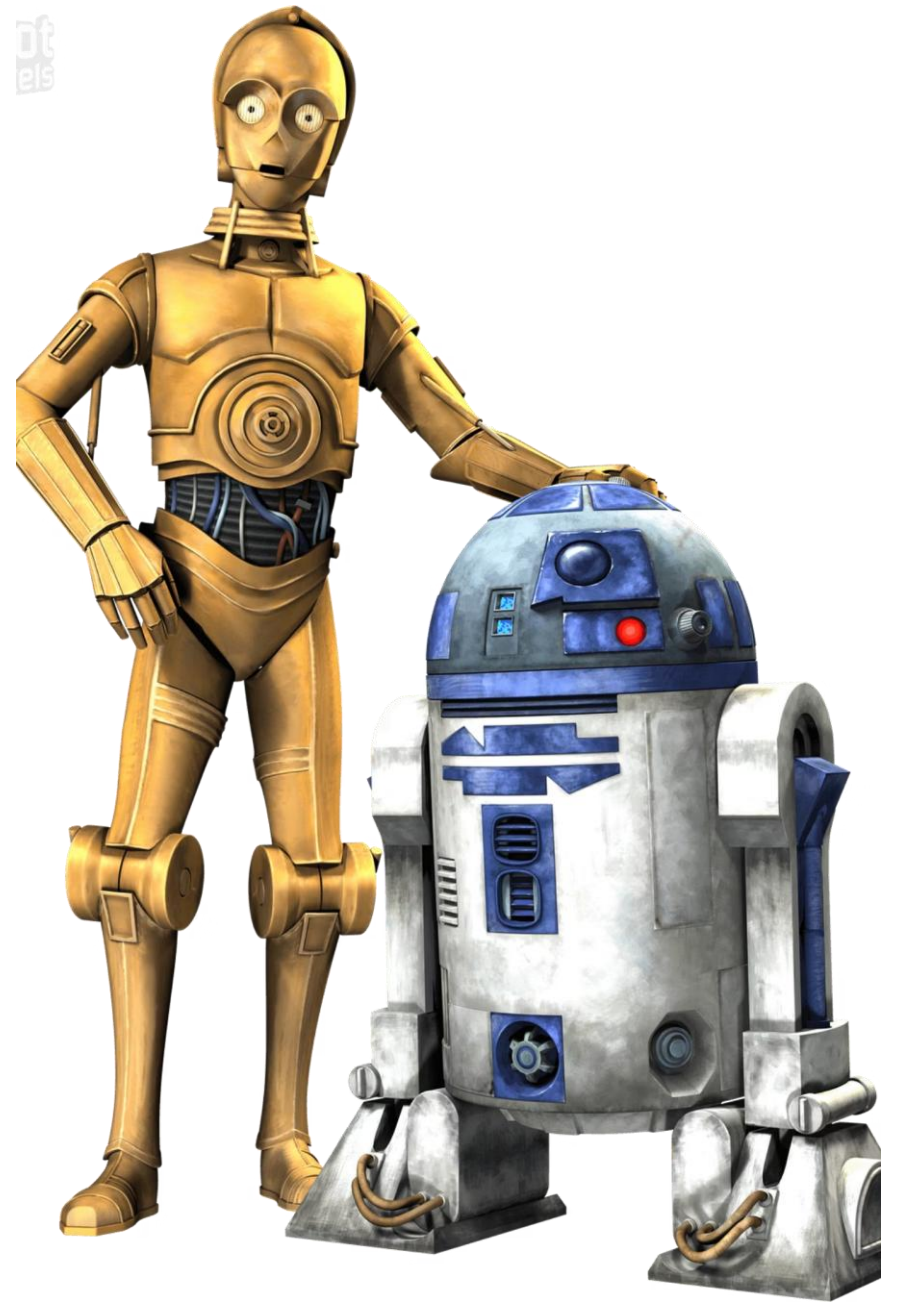
Document your assumptions and your reasons for them. Put years in columns and items in rows throughout the Excel workbook and PowerPoint deck.

Microsoft's
Vibe Working

Powered by Claude

Agent Mode
in Excel

Chatbots and Agents



Chatbot Elements

1 **User Interface**
The front-end interface where users interact with the system

2 **API Connection**
Secure connection to Large Language Model (OpenAI, Anthropic, Google, etc.)

3 **System Prompt**
Instructions that define the chatbot's behavior (on top of API system prompt)

4 **Document Retrieval**
Optional capability to access specific documents or knowledge bases

5 **Tool Integration**
Optional connections to external tools, databases, or APIs

RAG: Retrieval Augmented Generation

Document Storage

Store relevant documents in a searchable database or vector store

Relevance Matching

Choose most relevant document chunks using vector similarity algorithms

Context Integration

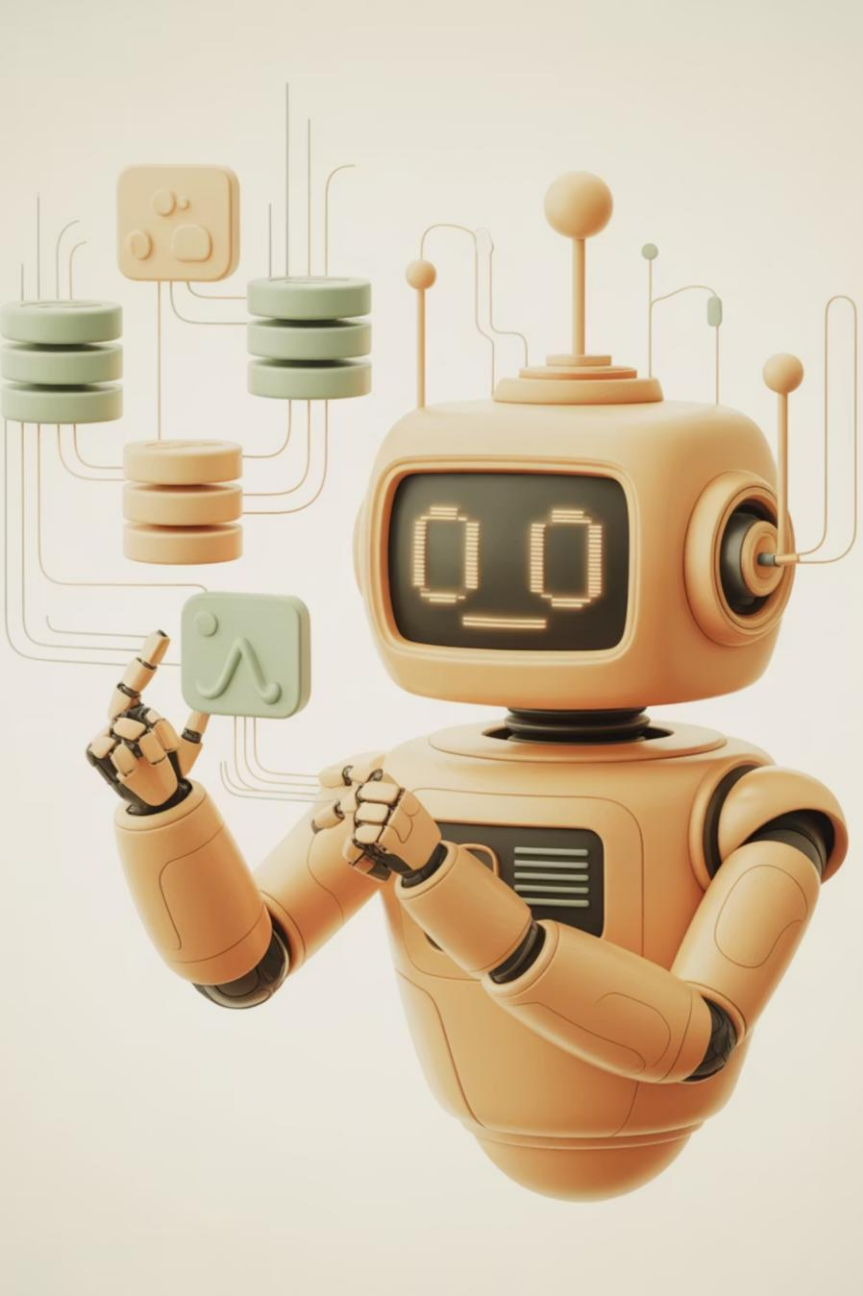
Send relevant chunks to LLM along with user prompt and system prompt

RAG is a key element of many corporate AI implementations.

A Vibe-Coded RAG Chatbot

Visit chat.derivative-securities.org





Agent = Enhanced Chatbot

Definition

An AI agent is a chatbot equipped with tools to perform actions beyond conversation.

Database Integration

Database tools are particularly valuable for chatbots in financial applications.

Claude for Financial Services

Anthropic: Through data providers, Claude for Financial Services has real-time access to comprehensive financial information.



Data Platforms

Box, Daloopa,
Databricks



Financial Data

FactSet, Morningstar,
Palantir



Market Intelligence

PitchBook, S&P
Global, Snowflake

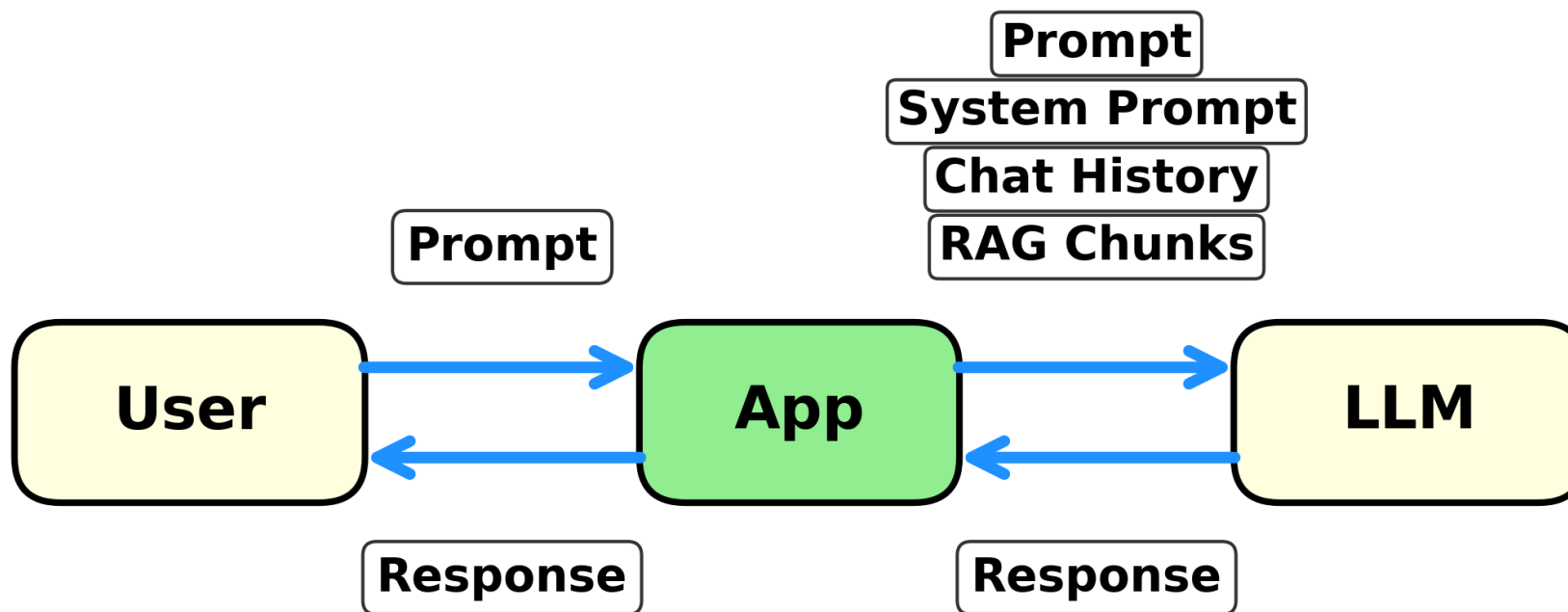
Why MBAs Should Learn about Agents

Fortune, September 14, 2025:

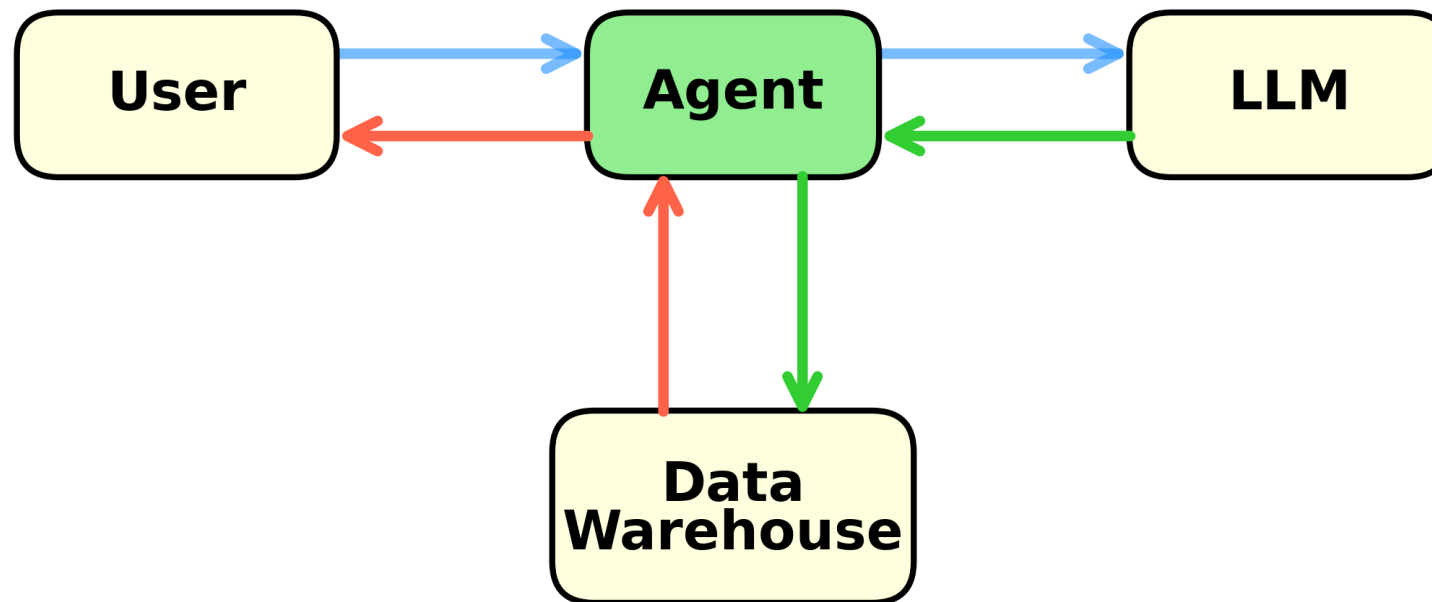
PromptQL, an enterprise AI platform created by San Francisco-based developer tooling company Hasura, is doling out **\$900-per-hour** wages to its engineers tasked with building and deploying AI agents to analyze internal company data using large language models (LLMs).

Tanmai Gopal, PromptQL's cofounder and CEO, said '**MBA types** ... are very strategic thinkers, and they're smart people, but they **don't have an intuition for what AI can do.**'

Chatbot Architecture



Database Agent



Could attach more tools. For example, Python to analyze data. Then Output → User



A Vibe-Coded Database Agent

[Visit data-portal.rice-business.org](https://data-portal.rice-business.org)

Example prompt:

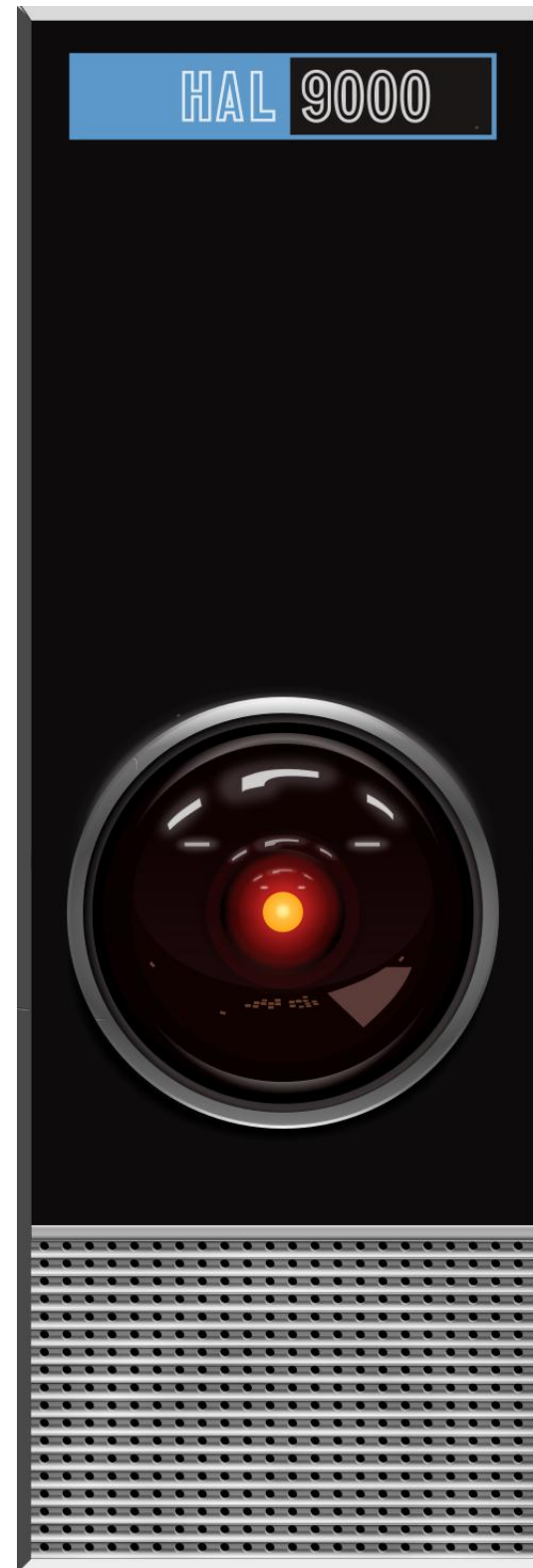
Get TSLA's ROE by quarter on a trailing 4-quarters basis for Q1 2020 through Q3 2025 from the Rice stock database.

Past, Present, & Future of UI's

- **Terminal Era:** Unix, MS-DOS, Linux, ...
- **Graphical User Interface:** Point & click
- **Chat:** AI agent does everything. Agents will change the way people use computers.



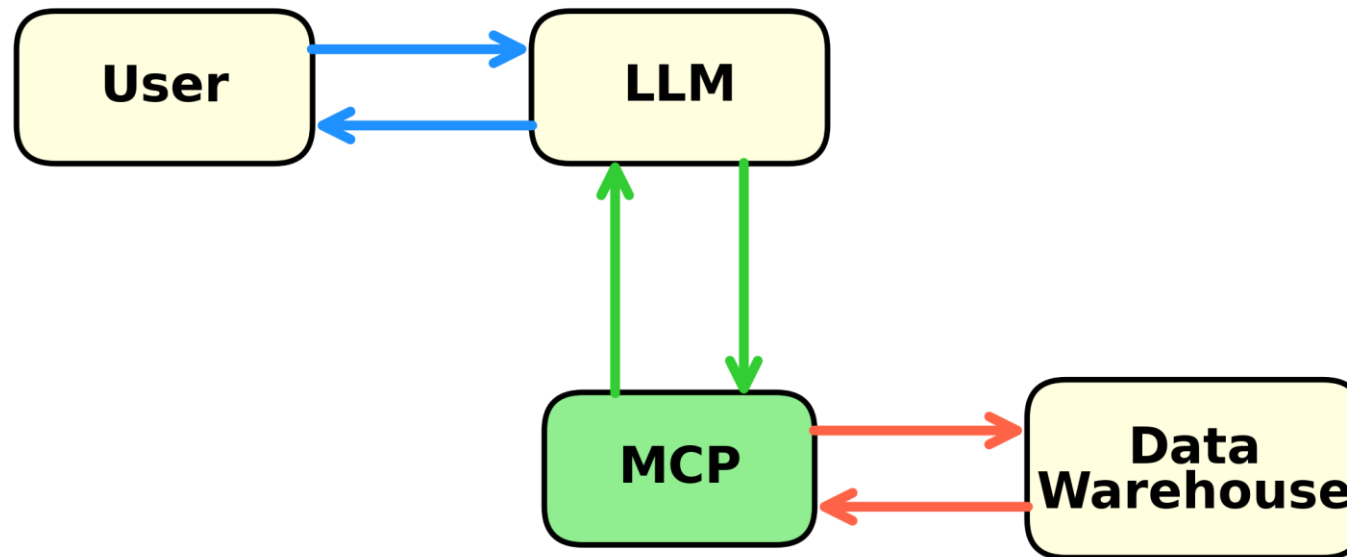
Or Soon This





BY ANTHROPIC

Model Context Protocol (Nov, 2024)



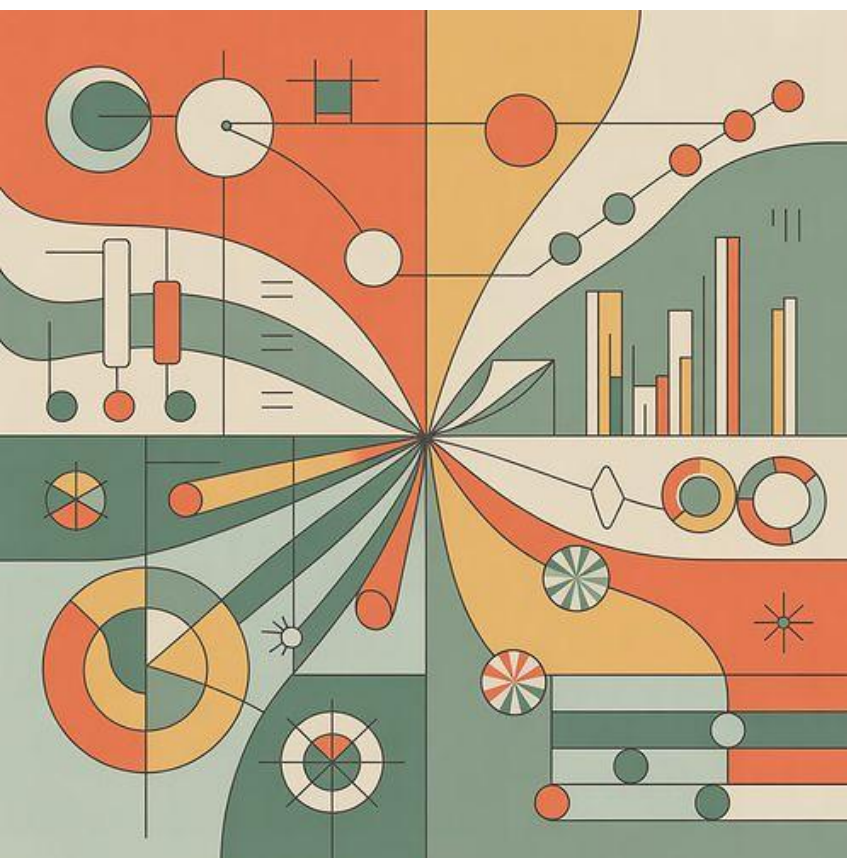
User chats directly with Claude

Agent logic embedded in MCP server (local or cloud) and instructions to LLM

User experience is simplified. Use Claude for everything.

OpenAI and others have adopted MCP.

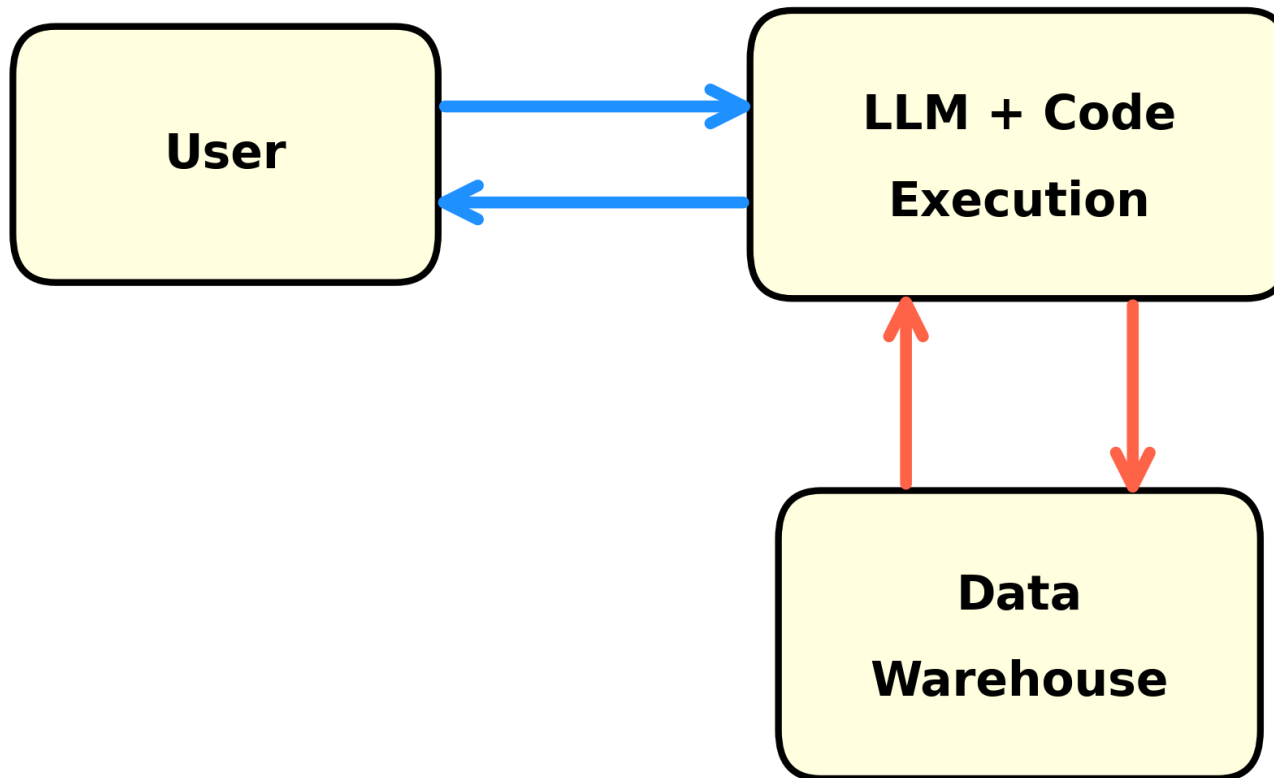
Vibe Coded MCP Database Server



I built an MCP server for the Rice database and connected it to my Claude Desktop.

Claude Prompt: Get TSLA's ROE by quarter on a trailing 4-quarters basis for Q1 2020 through Q3 2025 from the Rice stock database **and plot it.**

Claude Skills (10-16-2025)



User chats with Claude as with MCP

Agent logic embedded in Skill – instructions that Claude reads when needed + Python scripts it can execute

Developer experience is simplified

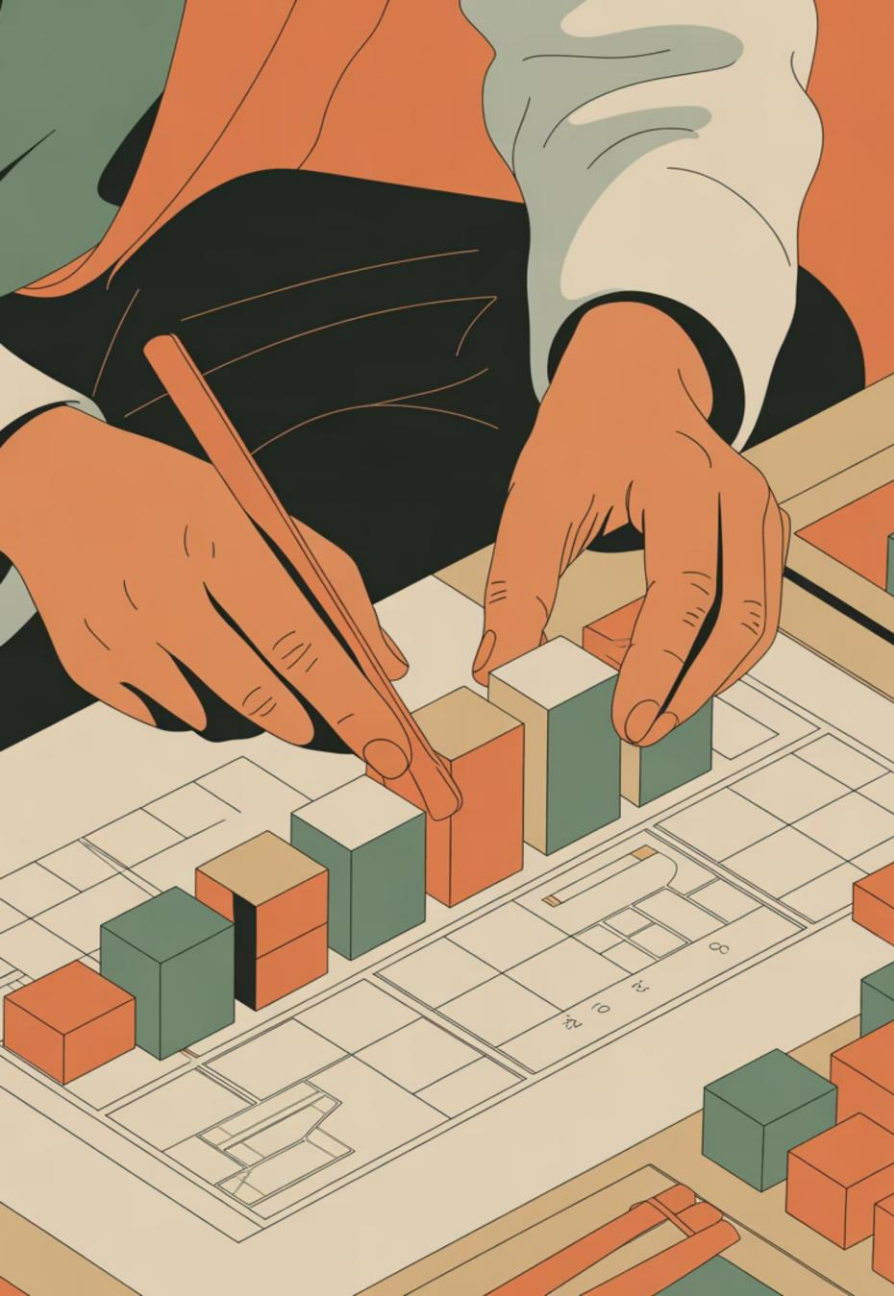
Claude has a Skill for creating Skills!



OpenAI



ChatGPT



OpenAI's Apps, Agent Builder, & Agent SDK (10-6-2025)

- Built on MCP
- OpenAI apps: ChatGPT becomes the user interface. **Goal is to be the new Apple App Store.**
- Launched with Coursera, Expedia, Spotify, Zillow, ...
- Tools for building apps:
 - Agent Builder: no-code platform
 - Agent SDK: code platform



Previous Disruptive Technologies

Handheld Calculators

The Concerns

Educators worried students wouldn't learn fundamental arithmetic operations

The Reality

Studies haven't found significant negative effects on basic math skills. However, primary schools still provide practice and assessments without calculators.

Some hoped schools could move beyond calculation to teach deeper mathematical concepts. That didn't happen either.



Personal Computers & Spreadsheets



We added courses on how to use spreadsheets

We updated existing courses to use spreadsheets

We assess how well students can use spreadsheets.

Teaching how to do finance in spreadsheets is an important educational goal.

Why Spreadsheets Succeeded

-  **Business Relevance**
Spreadsheets are extensively used in real business environments
-  **Educational Value**
Ideal for teaching the logic of valuation and capital budgeting concepts
-  **Tool Evolution**
Spreadsheets don't automatically do what we want students to learn

AI: A Different Challenge

✓ Business Usage

AI is increasingly used across business functions

✓ Educational Tool

AI provides excellent tutoring and learning support

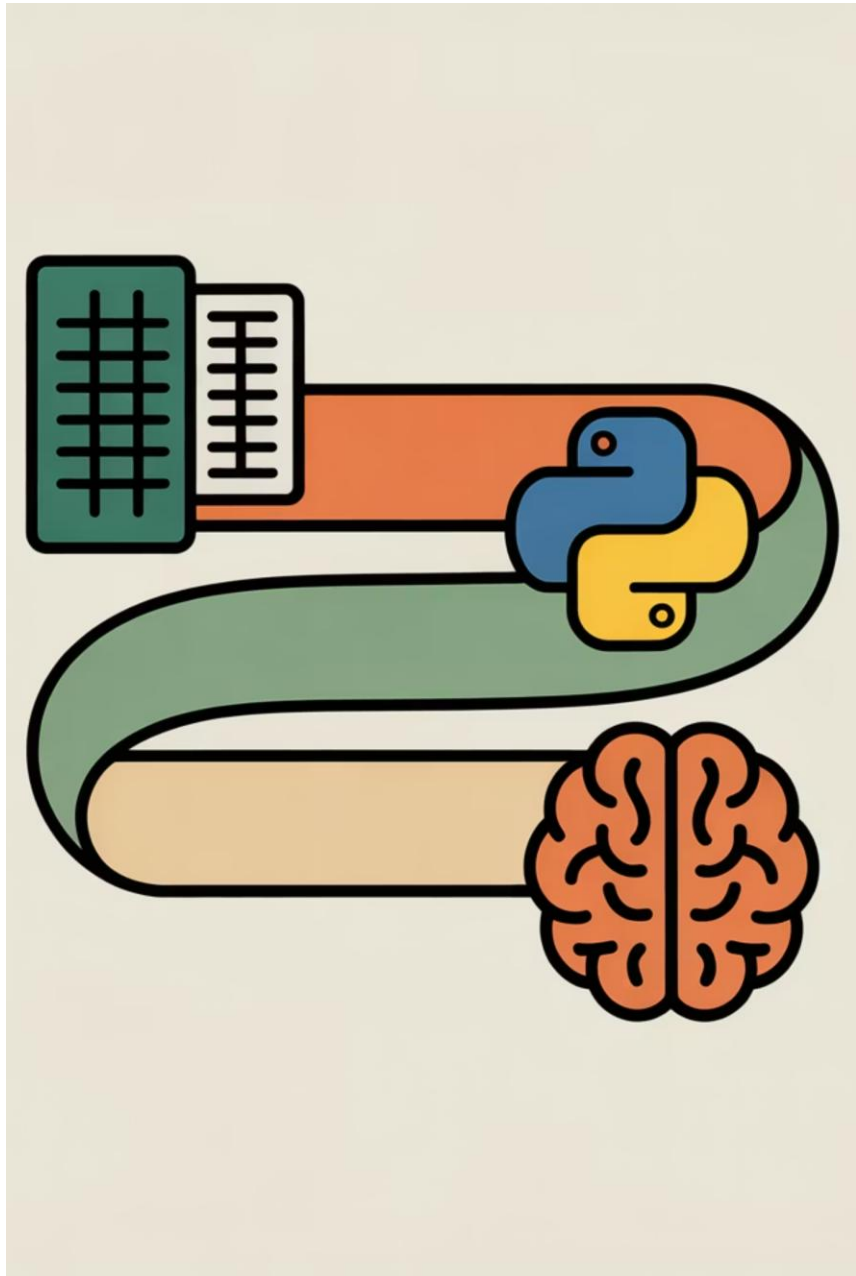
✗ Automation Risk

AI can automatically perform many tasks we previously taught manually

AI is more like calculators than spreadsheets. We face similar concerns about skill acquisition. Like primary schools with arithmetic, we'll need practice and assessments without AI to ensure fundamental competencies.

AI in Finance Courses





The New Skills Landscape

Brookings Institute, 2025:

As AI models begin to handle underwriting, compliance, and asset allocation, the traditional architecture of financial work is undergoing a fundamental shift.

As job descriptions evolve, so does the definition of financial talent. **Excel is no longer a differentiator. Python is fast becoming the new Excel.**

But technical skills alone will not cut it. The **most in demand profiles today are those that speak both AI and finance.**

A New Teaching Sequence

1

Concepts First

Begin with fundamental concepts using slides or whiteboard as usual

2

Traditional Tool Implementation (Maybe?)

Demonstrate implementation in traditional tools (usually Excel) as usual

3

AI Tool Implementation

New step: **Apply concepts using chatbots and AI agents**

Traditional Approach

Teach Excel implementation → students build Excel models

Additional Piece

Teach AI implementation → **students conduct AI chats or build AI apps**

Do We Need the Excel Step?

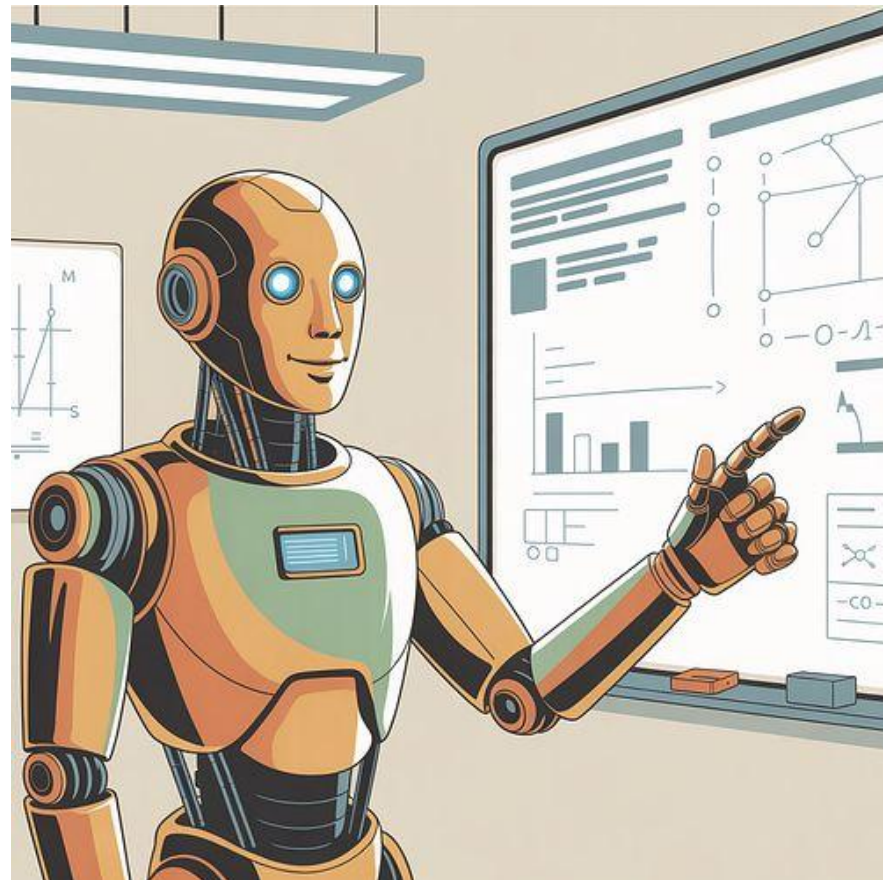
Linear regression

Simulation

Mean-variance Optimization

Option valuation

DCF analysis



Do We Need the Excel Step?



Linear regression



Simulation



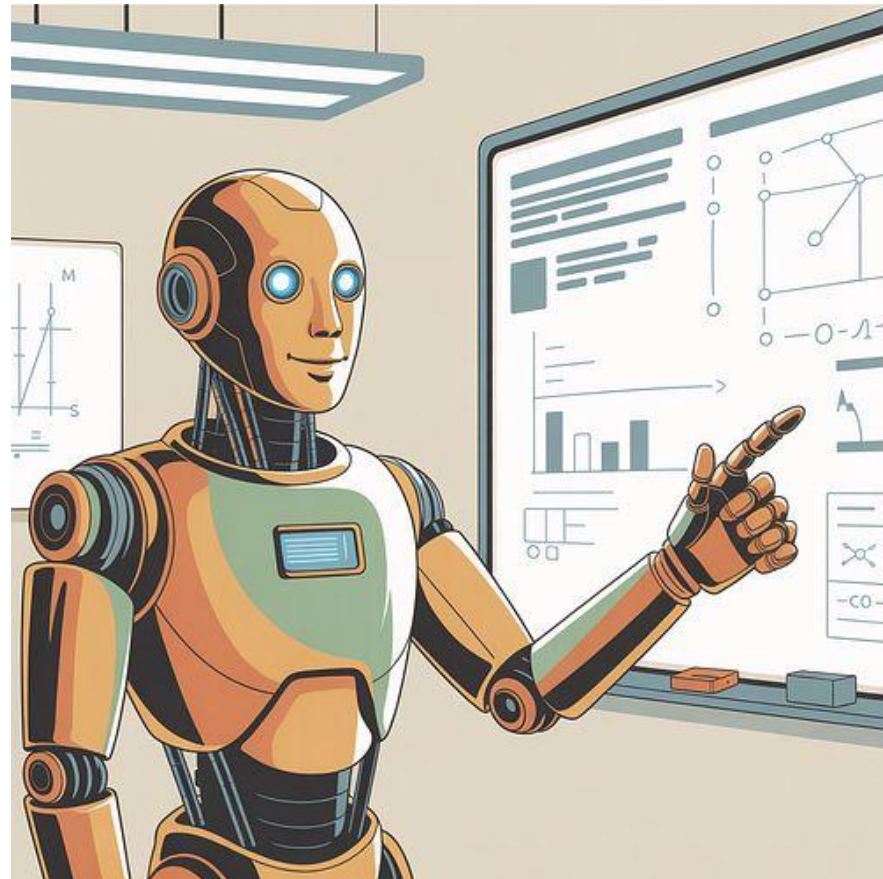
Mean-variance Optimization



Option valuation



DCF analysis



How Can We Assess – in All Courses?

Chat Assignments

"Turn in your chat, its output, and your evaluation of the output"

Oral Assessments

Cold calling, grading mostly on presentations with tough questions, even oral exams?

Exams Without AI

In-class assessments ensuring students understand fundamental concepts independently

Exams With AI

More ambitious assessments leveraging AI assistance - complex cases can become exams with AI support



AI as Teaching Partner

Learning from Chatbots

AI can be a highly effective tutor, and we should leverage it in our teaching.

Replace Reading Assignments

Ask students to chat with chatbots instead of traditional reading assignments



Pre-Class or In-Class

Assign pre-class or in class before slides and class discussion.

Assignment: "Ask your chatbot to teach you about [topic]. Tell it to ask you questions to ensure you are understanding."

If Chatbot Reliability is a Concern

Use RAG

Build custom systems (could use **Google's Notebook LM** or Custom GPT) for specific source control

Build with AI Reports

Can have AI create a comprehensive report, then edit if needed and use for RAG or in system prompt

Reliability Reality

LLMs today are significantly more reliable than earlier versions - hallucinations have diminished

NotebookLM Example

Step 1:

I asked ChatGPT to prepare a detailed report on binomial option pricing in markdown format. I could have used a different chatbot.

Link to ChatGPT conversation:

<https://chatgpt.com/share/68e7cbf4-013c-8006-89bb-1166720463ff>

Step 2:

I uploaded the markdown file to NotebookLM and asked it to generate flash cards and a video overview.

Link to NotebookLM notebook (Rice only):

<https://notebooklm.google.com/notebook/7e3e40f0-a937-46cb-8eea-7c713bdcb8d4>

Link to NotebookLM video:

https://genai4finance.kerryback.com/Binomial_Option_Model.mp4

Effect of AI on Junior Employment

Harvard economists' study:

AI is reducing
• employment of junior workers but not senior workers

[Lichtinger and Maasoum](#) (20,000+ SSRN downloads in one month)

JPMorgan quote:

"Many, many of the
• processes underneath are being done by AI."

Junior professors?

[Scott Latham, UMass, Chronicle of Higher Education:](#)

[Are you Ready for the AI University?](#)



Deep Dive into Claude Code

Claude Code

Part of a new generation of terminal-based coding agents, along with OpenAI Codex and Gemini CLI.

It integrates well with popular development environments like VS Code.

Using standard Anthropic models, it is optimized for development tasks.



Requires Claude Pro subscription

Varieties of Claude



Claude.ai

Interact with Claude directly through its intuitive web interface for general tasks and conversations.



Claude Code in Terminal

Integrate Claude's coding capabilities into your command-line workflow for quick code generation and debugging.



Claude Desktop

Access Claude as a dedicated desktop application for enhanced productivity and offline capabilities.



Claude Code in VS Code

Leverage Claude's advanced code assistance directly within your VS Code development environment.

Claude Code's Capabilities



File Operations

Create, edit, copy, move, and delete files



Python Development

Write, edit, and execute **Python scripts** and **Jupyter notebooks**



Document Creation

Generate and edit **LaTeX** documents. Generate **Word** docs, **PowerPoint** presentations, and **Excel** workbooks



Deployment

Create **GitHub** repositories and **deploy apps** on hosting platforms like Hugging Face Spaces, Streamlit Community Cloud, Koyeb, and Render

Claude Code Example 1: Creating Apps

Create a Streamlit app that allows a user to input (i) years to retirement, (ii) current investment account balance, (iii) planned annual savings, (iv) expected rate of return, and (v) standard deviation of rate of return.

Simulate 1,000 possible lifetimes and display statistics and a probability density curve for the account balance at retirement.

Run the app and tunnel it using my ngrok token stored at C:\users\Kerry\.env.



Claude Code Example 2: Skills

Claude Prompt 1: Research how to install Skills and then **install the xlsx and docx Skills**.

Claude Prompt 2: Create a **mean-variance-optimization Skill**.

It should prompt the user for the number of assets and their expected returns, standard deviations and correlations (the minimum number of correlations needed to compute the correlation matrix), and for the risk-free rate.

Then calculate the tangency portfolio, allowing short sales.

Then create a Word doc containing the relevant data, the tangency portfolio, and your explanation of why the tangency portfolio is what it is. Save the doc in the user's working directory.

Claude Code Example 2: Skills

Claude Prompt 1: Research how to install Skills and then **install the xlsx and docx Skills**.

Invoke the new Skill and use it.

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Claude Code Example 2: Skills

Claude Prompt 1: Research how to install Skills and then **install the xlsx and docx Skills**.

Invoke the new Skill and use it.

Claude Prompt 3: Review the conversation and improve the Skill.

Claude Prompt 2: Create a **mean-variance-optimization Skill**.

It should prompt the user for the number of assets and their expected returns, standard deviations and correlations (the minimum number of correlations needed to compute the correlation matrix), and for the risk-free rate.

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Claude Summary



Finance courses should leverage **Claude Desktop** for an intuitive interface and practical applications.

Courses on **AI in finance** should utilize **Claude Code** for model building.

When Should Students Learn Claude

(Or Whatever is the Best AI Tool at the Time)

- First term as a tools course (or in Data Analysis, for example)
- Last term as an advanced elective



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HBS licensed Julius.ai to use in Data Analysis course.



Domain Knowledge will be More Important than Ever

- AI will level the playing field for tool skills (assuming students can use AI).
- That will make domain knowledge more important than ever as a differentiator.
- I think that means **we should teach even more cases.**
- But maybe not as group projects. One Person + AI $\geq$ Two People





Some Additional Resources

Scan the QR code to access
genai4finance.kerryback.com